

ADAPTIVE FACTOR ALLOCATION VIA MACHINE LEARNING REGIME DETECTION *

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Abstract. Equity factor payoffs are strongly time-varying, raising the practical question of whether factor exposure can be adapted across market regimes using a simple, pre-specified, implementable rule rather than discretionary timing. We study adaptive factor allocation via machine-learning-based regime classification in a long-only U.S. equity setting. Using 25 fundamental characteristics, we construct investable signal portfolios and aggregate them into two tradable sleeves: an Offensive basket (growth-oriented signals) and a Defensive basket (quality and conservative signals). Market regimes are defined from an EWMA volatility proxy computed on an investable benchmark and mapped into a binary high/low state. We train multiple supervised classifiers under a common feature set and evaluate them in a walk-forward design with annual refits at June formation dates and a fixed July–June holding horizon. The out-of-sample evaluation spans July 1980–June 2024 and enforces strict point-in-time constraints and explicit timing alignment to preclude look-ahead bias. Performance is reported gross and net of turnover-scaled transaction costs, and risk-adjusted results are assessed under alternative factor models with HAC inference and multiple-testing diagnostics. We find that several regime-switching configurations improve net performance relative to the market benchmark with an economically modest gross-to-net wedge under annual implementation. However, incremental gains relative to transparent buy-and-hold allocations (including a 50/50 Offensive/Defensive benchmark) are typically modest, benchmark-dependent, and sensitive to the risk-adjustment model. Overall, machine learning can add value for regime-based factor allocation in some configurations, but the effect is heterogeneous and implementation frictions are first-order for net performance.

Key words. machine learning; regime detection; factor investing; asset allocation; volatility regimes; long-only portfolios; transaction costs; out-of-sample evaluation.

AMS subject classifications. 62P20, 91G10, 62M10, 68T05

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1. Introduction. Equity factor payoffs are strongly time-varying. For long-only investors, this creates a practical allocation problem: static exposures to “offensive” (growth- and momentum-tilted) versus “defensive” (quality- and conservatism-tilted) factor sleeves may be easy to hold, but can be poorly matched to changing market environments. This paper asks whether a simple, implementable rule can adapt factor exposure across regimes without discretionary timing.

Machine-learning-based timing is appealing, but two empirical pitfalls are recurrent. First, regime detection is highly sensitive to timing and alignment: labels, features, and formation dates live on different calendars, and small lookahead mistakes can create large apparent gains. Second, exploring many specifications mechanically produces dispersion, and ex-post selection can overstate the incremental value of the learning step. The relevant question is therefore whether machine learning adds value *net of costs* and relative to simple buy-and-hold benchmarks under a strictly aligned protocol.

We study adaptive factor allocation via regime classification in a long-only U.S. equity setting. Starting from 25 firm-characteristics signals, we form investable signal portfolios and aggregate them into two tradable sleeves: an *Offensive* basket and a *Defensive* basket. Regimes are defined by an EWMA volatility proxy on an investable benchmark and converted into a binary high/low state. We compare five basket constructions and five classifier families (25 strategies) in a walk-forward design with annual refits at June formation dates and a fixed July–June holding horizon, evaluated out of sample from July 1980 to June 2024. Performance is reported gross and net of transaction costs, and risk-adjusted results are assessed under alternative factor models with HAC inference; incremental value is benchmarked against simple allocations including a fixed 50/50 Offensive–Defensive portfolio.

The results point to a qualified conclusion. Dynamic beta baskets tend to deliver stronger net performance than the fixed economic basket, but the incremental timing component is heterogeneous and benchmark-dependent: alphas are often positive, yet statistical support is uneven, and incremental gains relative to buy-and-hold benchmarks are typically modest and sensitive to the risk model. Trading frictions matter for realizability, although under annual June implementation the gross-to-net wedge remains economically limited.

The remainder of the paper is organized as follows. Section 2 states the research question and positions the contribution in the related literature. Section 3 presents signal construction, basket definitions, regime labeling, and the walk-forward backtest protocol. Section 4 describes the data sources and investable universe. Section 5 summarizes implementation details and anti-lookahead checks. Section 6 reports out-of-sample performance and robustness. Section 7 concludes.

2. Research question and relevant literature.

2.1. Research question. Equity factor premia exhibit substantial time variation, creating a tension between simple static factor mixes and adaptive allocations that respond to changing market conditions. This capstone asks whether supervised machine learning can deliver an implementable improvement in a long-only factor-allocation setting by switching between Offensive and Defensive factor sleeves across market regimes.

Formally, the main research question is:

Can ML-based regime detection improve the net (after-cost) performance of a long-only factor allocation strategy relative to transparent buy-and-hold benchmarks, under a strictly point-in-time and out-of-sample evaluation design?

We operationalize this question through three testable sub-questions:

1. **Economic value net of costs:** Does regime switching improve *net* performance *relative to buy-and-hold benchmarks* once turnover and transaction costs are accounted for?
 2. **Risk-adjusted performance:** Are improvements robust after controlling for standard risk factors using HAC inference?
 3. **Robustness and selection risk:** Are gains broad across model families and portfolio constructions, or concentrated in a small subset of specifications?
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2.2. Relevant literature and positioning. Our setting relates to three strands of work. First, it builds on the large factor-investing literature and the debate on factor proliferation and replicability. “Factor zoo” and data-mining concerns motivate conservative inference and robustness when proposing incremental improvements.(5; 8; 4; 9) In this spirit, we emphasize out-of-sample evaluation, cross-specification dispersion, and risk-adjusted tests rather than highlighting a single ex-post “winner”.

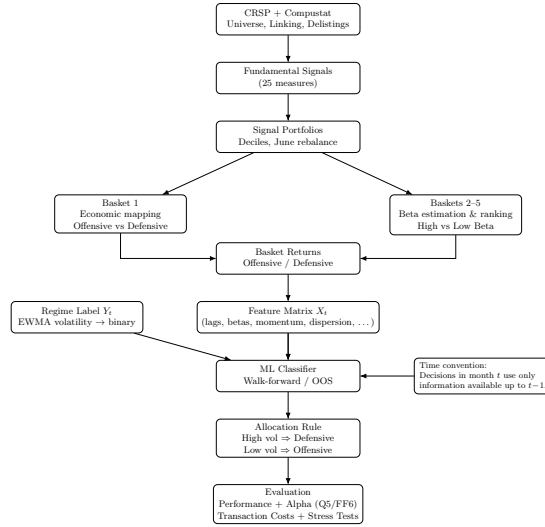
Second, our approach connects to evidence that market conditions—in particular volatility and risk appetite—proxy for regimes that affect the relative performance of growth/offensive versus quality/defensive exposures. We define regimes using a RiskMetrics-style EWMA volatility proxy.(12) Our Defensive sleeve is conceptually related to quality-oriented factors, including QMJ.(1) Related cross-market evidence on style premia (e.g., value and momentum) further motivates regime-dependent allocations across broad factor exposures.(2)

Third, any realistic assessment of timing strategies must incorporate implementation frictions. Trading costs and turnover materially affect the net profitability of anomaly- and factor-based strategies, motivating our focus on net performance and explicit turnover accounting.(11; 6; 7)

2.3. Contribution. Relative to this literature, our contribution is empirical and methodological. We evaluate ML-based regime allocation in a long-only, investable U.S. equity universe (CRSP/Compustat) under strict point-in-time construction and a walk-forward protocol with a common out-of-sample window. We report gross and net results, assess risk-adjusted alphas under multiple factor specifications, and document how conclusions depend on model class, portfolio construction, and trading frictions.

3. Methodology.

3.1. Pipeline Overview. We follow an annual June walk-forward protocol with a fixed July–June holding period, using only information available up to $t - 1$ (Tables 9–10).

FIG. 1. *Macro architecture of the pipeline*

Note: Data construction, signal portfolios, basket formation, regime labelling and feature engineering, machine learning classification, allocation, and evaluation.

3.2. Signal portfolios. Following the June-formation / July–June holding convention (3), we form annual characteristic-sorted portfolios for each signal using NYSE breakpoints. For each June y , we keep a single implementable long-only decile (top or bottom depending on the signal direction), equal-weighted and held from July y to June $y+1$. Signal definitions are in Appendix B.1 and the universe is summarized in Table 6.

3.3. Offensive and Defensive sleeves. We use the same 25 long-only signal portfolios and form two sleeves (Offensive/Defensive) under five mapping rules. Basket 1 applies a fixed ex-ante economic mapping (Appendix B.1, Table 6). Baskets 2–5 use a dynamic beta-based mapping: signals are classified into High-Beta (Offensive) vs. Low-Beta (Defensive) using lagged market-beta estimates (expanding, rolling-60m, EWMA, Kalman; Appendix B.2). Within each sleeve, returns are equal-weighted across signals; the allocator tilts between sleeves at the annual decision date.

3.4. Regime label. The supervised target is a binary volatility regime built from an EWMA volatility proxy on an investable market benchmark:

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{\text{mkt},t}^2, \quad \lambda = 0.94(12).$$

We set $Y_t = 1$ (high-vol) if σ_{t-1} exceeds the expanding median of past volatilities $\{\sigma_\tau\}_{\tau < t-2}$, and $Y_t = 0$ otherwise, ensuring the label uses only pre- t information.

3.5. Features. We construct a monthly feature vector X_t from lagged sleeve/signal information (returns, momentum, dispersion, beta dynamics, cross-sectional aggregates, and interactions). All features are lagged by one month; the full list (35 features) is reported in Appendix B.3 (Table 7).

3.6. ML models and walk-forward evaluation. We evaluate four classifier families (logistic regression, random forest, gradient boosting/XGBoost, and an LSTM); hyperparameters are in Appendix B.4 (Table 8). Each June y , models are refit on an expanding training set ending before the decision date and produce one regime forecast for the subsequent July–June holding year (low-vol: Offensive; high-vol: Defensive). All models share the same out-of-sample window (July 1980–June 2024); initialization requirements and comparability are summarized in Tables 9–10.

3.7. Transaction costs. We report gross and net returns. Net returns deduct proportional transaction costs applied to realized turnover at the *annual June reconstitution* of the underlying long-only signal portfolios (June formation; July–June holding). Let $\tau_t = \frac{1}{2} \sum |\Delta w|$ denote one-way turnover and c_t the one-way cost rate (in bps). We use a time-varying one-way cost schedule $c_t \in \{50, 30, 20\}$ bps for 1980–1999, 2000–2009, and 2010–2024, respectively. Monthly net returns are

$$r_t^{net} = r_t^{gross} - \mathbb{1}\{t \in \text{June}\} \frac{2 \tau_t c_t}{10,000}.$$

The baseline charges holdings-level trading frictions only; a conservative sensitivity analysis additionally charges allocation-layer (inter-sleeve) switching costs. The time-varying cost schedule follows institutional estimates and reflects secular declines in trading frictions (11; 6; 7). Turnover decompositions and break-even transaction-cost headroom relative to the market and 50/50 benchmarks are reported in Appendix D.8 (Table 22).

3.8. Performance and inference. We report out-of-sample (i) portfolio performance (mean, volatility, Sharpe/Sortino, drawdowns) and (ii) classification metrics (accuracy, precision/recall, F1, ROC–AUC, Brier score). Risk-adjusted performance is assessed via time-series regressions on FF and Q-factor models with Newey–West HAC inference (13); multiple-testing (FDR) adjustments are provided in the appendix (5; 8).

4. Data. We study U.S. common stocks over July 1963 to June 2024. Monthly returns, prices, shares outstanding, exchange/share codes, and delisting information are obtained from CRSP. Annual accounting variables are from Compustat, merged to CRSP identifiers using the CRSP–Compustat Merged (CCM) link table. Benchmark factor returns (FF5/FF6/FF7 and Q5) are used only for risk-adjusted performance tests.

4.1. Datasets.

- **CRSP (monthly stock file):** Monthly returns and firm characteristics for U.S.-listed equities; we retain ordinary common shares only (share codes 10 and 11).
- **Compustat (fundamentals annual):** Firm-year financial statements used to compute the 25 fundamental characteristics.
- **CCM link table:** Mapping between CRSP PERMNO and Compustat GVKEY. When multiple links exist, we prioritize the most reliable CCM link types to enforce a one-to-one mapping; full details are provided in the appendix.
- **Factor benchmarks:** Fama–French factors (including RF) and the Q5 factors are used only as controls in time-series regressions for alpha and inference.

4.2. Sample construction.

Point-in-time convention. To eliminate look-ahead bias, portfolios formed in June of year t use fundamentals from fiscal year $t-1$ (with a conservative reporting lag) and market equity measured at end-May t , consistent with the June formation / July–June holding convention. This ensures that all signal inputs used at formation are observable prior to implementation.

Delistings. We use CRSP delisting returns (DLRET) when available. When performance-related delistings have missing DLRET, we apply a stochastic imputation designed to avoid the upward bias documented in the literature (Shumway, 1997)(14); implementation details and parameter choices are reported in Appendix C.2. Delisting-month returns are adjusted as

$$r_{i,t}^{adj} = (1 + r_{i,t})(1 + \text{DLRET}_{i,t}) - 1,$$

and firms permanently exit the panel after the delisting month.

Investable universe filters. We apply the following sequential filters at each annual June formation:

1. **Common stocks:** retain CRSP share codes 10 and 11.
2. **Major exchanges:** retain NYSE/AMEX/NASDAQ listings.
3. **Price screen:** require price $\geq$ \$5 at formation.
4. **All-but-microcaps:** exclude stocks below the NYSE 20th-percentile market-cap breakpoint (computed within NYSE and applied to the full universe).
5. **Signal availability:** for each signal, require non-missing accounting inputs needed to compute the characteristic (signal-specific coverage is reported in Appendix C.3).

Table 11 in Appendix C.1 summarizes the impact of these filters on the average cross-section size across subperiods. The final investable universe averages roughly 2,348 firms per June formation, focusing the analysis on a large- and mid-cap segment relevant for institutional implementation rather than microcap-driven paper profits.

5. Implementation. All data processing, feature engineering, model training, and backtesting are implemented in Python (Jupyter). We rely on `pandas/numpy` for data handling, `scikit-learn` and `xgboost` for ML estimation (`tensorflow/keras` for the LSTM), and `statsmodels` for econometric inference.

5.1. Pipeline and anti-lookahead checks. The code mirrors the four-stage pipeline in Section 3: (i) build the CRSP–Compustat investable universe and 25 long-only signal portfolios; (ii) form Offensive/Defensive sleeves under the five basket rules; (iii) compute the monthly regime label and the 35-feature matrix with a global one-month information lag; (iv) refit models each June in an expanding walk-forward design and evaluate out-of-sample performance over July 1980–June 2024.

To enforce point-in-time validity, all objects used at time t depend only on information through $t-1$: betas are lagged ($\beta_{s,t-1}$), features are shifted by one month, and the volatility label uses returns through $t-1$ with an expanding-median threshold based only on past observations. Training windows are verified to end strictly before each June decision date (Table 9).

5.2. Reproducibility. We fix random seeds for all stochastic components (seed = 42). Signals, betas, features, and labels are generated once and reused across model

classes. Tables and figures are exported directly from notebooks to LaTeX to limit manual handling.

6. Results. This section presents out-of-sample results for ML-based regime switching between Offensive and Defensive sleeves, including gross vs. net performance, risk-adjusted alphas, subperiod stability, and prediction diagnostics. Strategy returns are net of costs unless stated otherwise; the NYSE benchmark is shown gross.

6.1. Performance Results.

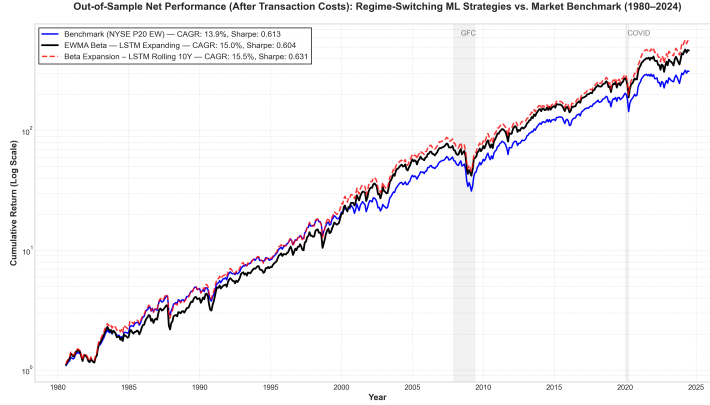
TABLE 1
Performance Comparison: Signal Power vs. Realizability (1980–2024)

Basket Strategy	Best ML Model	Ann. Ret. (%)	CAGR (%)	Vol. (%)	Sharpe	Sortino	Max DD (%)	DD Date
Panel A: Gross Returns (Signal Power)								
Beta Expansion	LSTM (Rolling 10Y)	16.85	15.88	19.96	0.649	0.933	-47.5	2009-02
Rolling Beta	LSTM (Rolling 10Y)	16.71	15.75	19.86	0.646	0.931	-44.8	2009-02
Kalman Filter Beta	LSTM (Rolling 10Y)	16.58	15.60	19.86	0.639	0.918	-44.9	2009-02
EWMA Beta	LSTM (Rolling 10Y)	16.55	15.57	19.86	0.638	0.913	-45.8	2009-02
Economic Basket	XGBoost	15.93	14.78	20.22	0.596	0.860	-49.0	2009-02
<i>NYSE Benchmark</i>	—	<i>14.71</i>	<i>13.95</i>	<i>17.66</i>	<i>0.613</i>	<i>0.846</i>	<i>-48.4</i>	<i>2009-02</i>
Panel B: Net Returns (Realizability)								
Beta Expansion	LSTM (Rolling 10Y)	16.47	15.45	19.97	0.631	0.906	-47.8	2009-02
Rolling Beta	LSTM (Rolling 10Y)	16.34	15.32	19.87	0.627	0.903	-45.1	2009-02
Kalman Filter Beta	LSTM (Rolling 10Y)	16.20	15.16	19.87	0.620	0.890	-45.2	2009-02
EWMA Beta	LSTM (Rolling 10Y)	16.16	15.13	19.86	0.618	0.885	-46.1	2009-02
Economic Basket	XGBoost	15.54	14.34	20.22	0.577	0.832	-49.3	2009-02
<i>NYSE Benchmark</i>	—	<i>14.71</i>	<i>13.95</i>	<i>17.66</i>	<i>0.613</i>	<i>0.846</i>	<i>-48.4</i>	<i>2009-02</i>

Notes: July 1980–June 2024. Panel A: gross; Panel B: net after June reconstitution transaction costs (turnover-scaled). “Best ML model” is selected by highest net Sharpe within each basket. Full grids are in Appendix D.1 (Table 13).

6.1.1. Out-of-Sample Performance: Signal Power vs. Realizability. Table 1 shows that the *dynamic beta* baskets outperform the fixed *Economic* basket on a risk-adjusted basis, both gross (Panel A) and net of costs (Panel B). The top specification (Beta Expansion + LSTM Rolling 10Y) delivers Sharpe ratios of 0.649 (gross) and 0.631 (net), exceeding the NYSE benchmark (0.613), while the gross–net wedge remains small (16.85% → 16.47% annualized return). Maximum drawdowns cluster in Feb. 2009 across all strategies, suggesting gains come from higher average risk-adjusted performance rather than crash-hedging. The limited cost drag is consistent with turnover concentrated at the annual June reconstitution (Appendix Tables 15 and 16).

6.1.2. Basket Building Blocks: Offensive vs. Defensive Sleeves. Table 14 reports unconditional sleeve performance (Offensive vs. Defensive), averaged across ML models. Defensive sleeves exhibit lower volatility and higher Sharpe ratios (about 0.62–0.66 gross and 0.62–0.63 net), while Offensive sleeves deliver similar long-run growth with higher volatility. This motivates the timing exercise: potential value should come from shifting toward Defensive exposure in high-volatility regimes while retaining Offensive exposure in benign states.

FIG. 2. *Out-of-Sample Net Cumulative Performance (After Transaction Costs): Best Regime-Switching ML Strategies vs. Market Benchmark (1980–2024)*

Notes: Log cumulative wealth (start=1) for the NYSE benchmark and the two best regime-switching ML strategies over July 1980–June 2024. ML series are net of transaction costs using the Frazzini-style June rebalancing convention (including inter-sleeve switching where applicable); the benchmark is shown gross for comparability with Table 1. Shaded bands mark the GFC (2007–2009) and COVID (2020).

6.1.3. Cumulative and Time-Varying Performance. Figure 2 shows that the two top regime-switching strategies outcompound the market benchmark net of transaction costs over 1980–2024, with performance differences remaining broadly stable through major stress episodes (GFC and COVID). Time-variation in risk-adjusted performance is documented in Appendix D.4 (Figure 4) via 60-month rolling Sharpe ratios and the corresponding Δ Sharpe.

6.2. Factor-Adjusted Performance (Alpha Analysis). We next assess whether the performance differences survive standard risk adjustments by estimating factor-model alphas on monthly *net* excess returns.

TABLE 2
Factor-Adjusted Performance of Buy&Hold Baskets and ML Strategies (Net)

Basket	Strategy	FF6			Q5			
		α (%)	t	R^2	α (%)	t	R^2	
Panel A: Buy&Hold Baskets (Net)								
Economic	(Offensive)	1.26		0.96	0.939	2.11	1.31	0.904
	(Defensive)	2.14	*	1.68	0.954	2.22	1.36	0.936
Beta Exp.	(Offensive)	2.82	*	1.94	0.951	3.00	1.59	0.934
	(Defensive)	1.28		1.17	0.945	1.80	1.22	0.911
Rolling β	(Offensive)	2.91	*	1.80	0.945	3.25	1.57	0.929
	(Defensive)	1.08		1.03	0.939	1.37	0.96	0.903
EWMA β	(Offensive)	2.62	*	1.70	0.945	2.98	1.51	0.932
	(Defensive)	1.33		1.19	0.937	1.64	1.12	0.900
Kalman β	(Offensive)	2.78	*	1.84	0.945	3.03	1.56	0.931
	(Defensive)	1.08		0.96	0.938	1.49	0.99	0.901
Panel B: Best ML Strategies by Basket (Net)								
Economic	XGB	1.73		1.31	0.941	2.14	1.32	0.908
Beta Exp.	RF	2.77	*	1.94	0.948	2.77	1.49	0.932
Rolling β	RF	3.14	**	2.02	0.940	2.99	1.49	0.926
EWMA β	RF	2.80	*	1.83	0.938	2.58	1.31	0.927
Kalman β	RF	2.82	*	1.91	0.940	2.59	1.35	0.927

Notes: Monthly *net* excess returns, 1980–2024. α annualized (% p.a.); Newey–West t (12 lags). FF6 and Q5 factor models. Panel B: best ML per basket (highest net Sharpe). ***/**/* denote 1/5/10%.

Table 2 reports factor-model alphas for the *buy-and-hold* sleeves (Panel A) and the corresponding *regime-switching* ML strategy for each basket (Panel B). Buy-and-hold alphas are generally positive but only weakly supported (Newey–West t -statistics mostly below 2). ML timing delivers similar or slightly higher FF6 alphas in a few cases (notably Rolling- β), but results are heterogeneous and typically weaker under Q5. Appendix Table 17 reports the full regression grid (FF5/FF6/FF7 and Q5) for all sleeves and all 25 ML strategies.

6.3. Incremental Alpha Relative to a 50/50 Buy-and-Hold Benchmark.

Table 3 reports $\Delta\alpha$ for the best model in each basket, along with summary statistics across all 25 ML specifications and a simple benchmark-robustness panel.

TABLE 3
Incremental Alpha: ML vs Buy&Hold 50/50 Benchmark

Basket	Best Model	FF6		Q5	
		$\Delta\alpha$ (%)	t	$\Delta\alpha$ (%)	t
Economic	XGB	0.03	0.12	-0.03	-0.10
Beta Exp.	RF	0.72	1.58	0.37	0.71
Rolling β	RF	1.14 *	1.92	0.68	0.90
EWMA β	RF	0.82	1.43	0.27	0.38
Kalman β	RF	0.89	1.61	0.33	0.46
<i>Average (all models)</i>	–	0.24	0.35	-0.19	-0.28
<i>Median (all models)</i>	–	0.21	0.30	-0.22	-0.46
Robustness: Alternative Benchmarks (FF6 only)					
Benchmark	Mean $\Delta\alpha$ (%)	% $\Delta\alpha > 0$ (FF6)	# Significant (FF6)		
50/50 (main)	0.24	64%	0/25		
Offensive only	-0.31	28%	0/25		
Defensive only	0.79	80%	1/25		

Notes: $\Delta\alpha$ is estimated from factor regressions on the *net* return differential $r_{ML,t}^{net} - r_{bench,t}^{net}$ (annualized, % p.a.). Main benchmark: ex-ante 50/50 (Buy&Hold Offensive + Buy&Hold Defensive). “Best model” is the highest net Sharpe within each basket. Costs are applied net-of-turnover under the annual June reconstitution using the same time-varying schedule for strategies and benchmarks. t -statistics are Newey–West (HAC, 12 lags). Sample: 1980–2024. ***/**/* denote 1/5/10%.

Table 3 isolates the timing component by differencing each ML strategy against a transparent 50/50 buy-and-hold benchmark. Incremental alphas are economically modest on average (mean $\Delta\alpha = 0.24\%$ p.a. under FF6) and rarely statistically strong: only the best specification reaches marginal significance under FF6, and inference weakens further under Q5. The robustness panel confirms benchmark sensitivity (higher $\Delta\alpha$ vs. a Defensive-only benchmark, weaker vs. Offensive-only), consistent with timing value being episodic rather than pervasive. The full 25-strategy grid (FF6 and Q5) with multiple-testing adjustments and additional robustness checks are reported in Appendix D.6 Tables 18–19; subperiod results are in Appendix Table 20.

6.4. Subperiod Analysis.

Table 21 in Appendix D.7 extends the subperiod exercise to the full grid of 25 ML strategies. Two regularities stand out. First, incremental performance is most consistently positive in the early sample (P1: 1980–1999), where dynamic beta baskets frequently exhibit economically meaningful $\Delta\alpha$ relative to the 50/50 benchmark. Second, results become markedly less stable thereafter: P2 (2000–2009) displays substantial dispersion across model classes within the same

basket (including both strongly positive and negative $\Delta\alpha$), while P3 (2010–2024) shows weaker and more mixed incremental performance overall. Taken together, the evidence supports an *episodic* interpretation of timing value and highlights that conclusions depend on both the market era and the chosen classifier/basket combination.

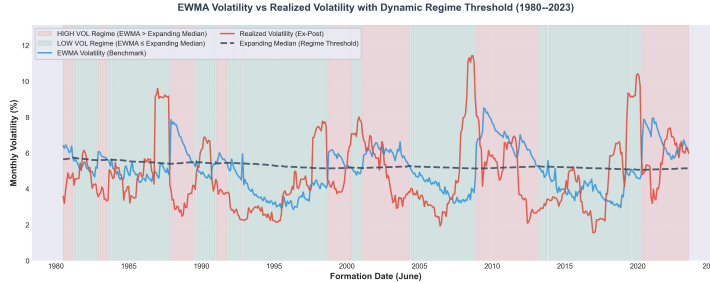
TABLE 4
Machine Learning Models: Regime Classification Performance (Annual June Decisions)

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Bal. Acc.	F1 Macro	Brier	Log Loss
Logistic	0.7273	0.6667	0.5000	0.5714	0.8393	0.6786	0.6857	0.1721	0.5192
Random Forest	0.7045	0.6667	0.3750	0.4800	0.7165	0.6339	0.6368	0.1967	0.5754
XGBoost	0.6591	0.5385	0.4375	0.4828	0.7321	0.6116	0.6143	0.2256	0.7140
LSTM (Exp.)	0.7273	0.7000	0.4375	0.5385	0.8192	0.6652	0.6725	0.1679	0.5006
LSTM (Roll.)	0.7273	0.6250	0.6250	0.6250	0.7076	0.7054	0.7054	0.2116	0.6492

Notes: Metrics are computed at the annual June formation date (one out-of-sample forecast per year for the subsequent July–June holding period). Walk-forward training uses an initial 1969–1979 window; out-of-sample evaluation is 1980–2023 ($N = 44$). Lower is better for Brier/log loss; higher is better otherwise. Bold indicates the best value.

6.5. Out-of-Sample Regime-Classification Performance (Annual June Decisions). Table 4 shows no clear winner: logistic regression has the highest ROC-AUC, LSTM (Rolling) the best precision–recall balance (F1), and LSTM (Exp.) the lowest calibration losses (Brier/log loss). Differences are modest, consistent with limited power from annual out-of-sample decisions ($N = 44$). We therefore also inspect the volatility label itself via the EWMA signal and its regime threshold (Figure 3).

FIG. 3. EWMA Volatility vs Realized Volatility with Dynamic Regime Threshold



Notes: EWMA volatility ($\lambda=0.94$, 60m window, MA(12)) on NYSE benchmark returns, formed each June using data through May. Realized volatility is the subsequent 12m (July–June). Regime threshold: expanding median of lagged EWMA. Sample: 44 June formations (1980–2023).

6.5.1. EWMA Volatility Signal and Dynamic Regime Threshold. Figure 3 shows that EWMA volatility is smooth and persistent, while realized volatility is spike-driven; the expanding-median threshold therefore produces an intentionally slow-moving high/low regime rule that captures broad stress episodes, consistent with annual implementation. Classification performance is evaluated at June decision dates ($N = 44$), while volatility-proxy calibration uses monthly forecast origins (1980m06–2023m06, $N = 517$); related calibration statistics are reported in Table 5.

6.6. Calibration of the Volatility Forecast. Table 5 shows that the EWMA proxy has weak volatility level-forecast accuracy (corr. = 0.083) but still provides modest information for binary high/low regime classification (Regime Acc. = 54.4%), which is the relevant object for the allocation rule.

TABLE 5
EWMA Volatility Forecast Calibration Quality (1980–2023)

Correlation	Regime Acc. (%)	Bias (%)	MAE (%)	RMSE (%)	N
0.083	54.4	-0.327	1.78	2.25	517

Notes: Monthly forecasts ($N = 517$, 1980m06–2023m06). $\hat{\sigma}_t$ is EWMA volatility ($\lambda = 0.94$, 60m window, MA(12)) formed with information through $t-1$ on NYSE benchmark returns; realized volatility is over $[t+1, t+12]$. *Regime Acc.:* agreement of forecast vs realized high/low states using their respective sample medians.

7. Conclusion. This section provides out-of-sample evidence (July 1980–June 2024) on whether machine-learning-based regime switching can improve adaptive allocation between Offensive and Defensive factor sleeves in a long-only setting. The headline result is economically positive but not uniform. Across basket constructions, regime-switching strategies built on dynamic beta mappings tend to deliver higher net Sharpe ratios and stronger long-run compounding than the market benchmark, and these gains remain largely intact after transaction costs (Table 1 and Figure 2). Consistent with the annual June implementation, the gross-to-net wedge is modest, which suggests that trading frictions do not mechanically dominate the strategy’s economics in this design.

A useful perspective is that the timing rule can only add value if the two sleeves it switches between are economically distinct. The sleeve decomposition confirms this structure: Offensive and Defensive legs exhibit broadly similar long-run growth but materially different risk profiles, with Defensive sleeves typically less volatile and stronger on downside-risk metrics (Table 14). This creates scope for regime timing, but it also implies that part of the performance differences can reflect basket construction and unconditional sleeve properties rather than the learning step itself.

Risk-adjusted tests qualify the interpretation. Time-series regressions generally yield positive alphas, yet statistical support is modest and sensitive to the chosen factor model: evidence is typically stronger under the Fama–French family and weaker under Q5 (Table 2). Incremental-alpha tests relative to a simple 50/50 buy-and-hold benchmark sharpen this point: $\Delta\alpha$ is economically modest on average and rarely statistically strong once standard adjustments and multiple-testing concerns are considered (Table 3 and Appendix Table 18). Subperiod evaluation (without re-optimizing the selected specifications) further indicates that incremental timing benefits are concentrated in earlier decades and attenuate thereafter, consistent with an episodic rather than time-invariant timing edge (Appendix D.7 –Tables 20).

Finally, predictive diagnostics align with the economic story. Regime classification quality at annual June decision dates is non-trivial but not decisive (Table 4), and the EWMA volatility proxy shows limited ability to forecast volatility *levels* while still providing modest information for coarse high/low regime separation (Figure 3 and Table 5). Taken together, the results suggest that ML-based regime timing can add value in some configurations—especially when basket construction creates meaningfully different sleeves and performance is evaluated net of costs—but that the incremental contribution of learning is heterogeneous, benchmark-dependent, and time-varying. The practical implication is that “ML timing” should be viewed as a conditional tool rather than a default upgrade: its value depends on disciplined timing alignment, transparent benchmarking, and explicit implementation assumptions, without which apparent gains are easy to overstate.

Appendix A. Helper tools.

The following helper tools were used in preparing this capstone:

- **ChatGPT (GPT-5.2 Thinking)**: editing for concision and SIAM formatting, restructuring sections to match the capstone requirements, and LaTeX assistance (e.g., figure/table placement, cross-references).
- **Python ecosystem**: data processing and analysis (polars/pandas/numpy), machine learning (scikit-learn, xgboost), and econometrics (statsmodels).

Appendix B. Methodology.

TABLE 6
Signal Universe and Basket Classification

Category	Signal Name	Leg	Economic Basket	Zoo ID
<i>Panel A: Profitability (5 signals)</i>				
	Gross Profitability	P10	Defensive	#88
	Operating Profitability	P10	Defensive	#95
	Return on Assets	P10	Defensive	#82
	Δ Asset Turnover	P10	Offensive	#72
	Δ Profit Margin	P10	Defensive	#74
<i>Panel B: Investment (6 signals)</i>				
	Return on Invested Capital	P10	Defensive	#70
	Asset Growth	P1	Offensive	#92
	Investment (CapEx)	P1	Offensive	#53
	CapEx + Inventory Growth	P1	Defensive	#81
	Net Share Issuance	P1	Defensive	#73
	CapEx Growth	P1	Offensive	#63
	Sales Growth	P1	Offensive	#24
<i>Panel C: Accruals & Quality (4 signals)</i>				
	Working Capital Accruals	P1	Defensive	#28
	Sales-Receivables Gap	P10	Defensive	#34
	Receivables Turnover	P10	Offensive	#17
	Operating Leverage	P10	Offensive	N/A
<i>Panel D: Liquidity & Solvency (4 signals)</i>				
	Cash Cushion	P10	Defensive	#86
	Debt Capacity	P10	Defensive	#71
	Book Leverage	P1	Defensive	#7
	Δ Current Ratio	P10	Defensive	#11
<i>Panel E: Operating Efficiency (5 signals)</i>				
	Asset Turnover	P10	Offensive	N/A
	Cash Productivity	P10	Offensive	#76
	Depreciation/PP&E	P1	Defensive	#18
	Δ Depreciation	P1	Defensive	#19
	Tax/Income Ratio	P10	Defensive	#56

Notes: This table presents the 25 fundamental signals employed across all five basket construction strategies. Signals are grouped into five economic categories following the asset pricing literature: Profitability, Investment, Accruals & Quality, Liquidity & Solvency, and Operating Efficiency. *Leg* indicates portfolio direction: P10 = buy top decile (high values are favorable), P1 = buy bottom decile (low values are favorable, contrarian). *Economic Basket* denotes the **fixed allocation** used in Basket 1 (Economic Classification): Offensive (9 signals) captures growth, expansion, and velocity metrics with procyclical behavior; Defensive (16 signals) aggregates quality, profitability, liquidity, and conservatism metrics providing downside protection. For Baskets 2–5 (Dynamic Beta Strategies), signals are **not** pre-assigned to baskets; instead, all 25 signals form a unified pool from which factors are dynamically allocated each month based on time-varying market beta estimates—top 50% highest beta $\rightarrow$ High Beta basket, bottom 50% $\rightarrow$ Low Beta basket. *Zoo ID* references Feng, Giglio, and Xiu (2020), “Taming the Factor Zoo,” *Journal of Finance*.

B.1. Basket Construction Details.

B.2. Beta estimation methods for dynamic baskets (Baskets 2–5). This appendix summarizes the market-beta estimators used to dynamically reclassify the 25 long-only signal portfolios into High-Beta (Offensive) and Low-Beta (Defensive) sleeves. Betas are computed from monthly excess returns $r_{s,t}^e = r_{s,t} - r_{f,t}$ and $r_{\text{mkt},t}^e = r_{\text{mkt},t} - r_{f,t}$, where $r_{f,t}$ is RF and $r_{\text{mkt},t}$ is the return on the investable market benchmark.

No-lookahead implementation. Classification for month t uses only information available through $t - 1$: signals are ranked by $\beta_{s,t-1}$, split at the median (top 50% High-Beta, bottom 50% Low-Beta), and the resulting sleeves are held in month t .

Basket 2: Expanding-window beta. Using all history from inception through $t - 1$,

$$\beta_{s,t}^{\text{EXP}} = \frac{\text{Cov}(r_s^e, r_{\text{mkt}}^e)_{[\text{start}:t-1]}}{\text{Var}(r_{\text{mkt}}^e)_{[\text{start}:t-1]}}.$$

Initialization: requires at least 60 monthly observations; first valid classification occurs in July 1968.

Basket 3: Rolling 60-month beta. Using a trailing 60-month window ending at $t - 1$,

$$\beta_{s,t}^{\text{ROLL}} = \frac{\text{Cov}(r_s^e, r_{\text{mkt}}^e)_{[t-59:t-1]}}{\text{Var}(r_{\text{mkt}}^e)_{[t-59:t-1]}}.$$

Initialization: 60 months; first valid classification occurs in July 1968.

Basket 4: EWMA beta ($\lambda = 0.94$). Exponentially decaying weights (RiskMetrics convention) (12):

$$\beta_{s,t}^{\text{EWMA}} = \frac{\sum_{\tau=1}^{t-1} \lambda^{(t-1)-\tau} r_{s,\tau}^e r_{\text{mkt},\tau}^e}{\sum_{\tau=1}^{t-1} \lambda^{(t-1)-\tau} (r_{\text{mkt},\tau}^e)^2}, \quad \lambda = 0.94.$$

Initialization: we require 60 months to stabilize the recursion; first valid classification occurs in July 1968.

Basket 5: Kalman-filter beta. We estimate a time-varying beta in a state-space model (10):

$$\begin{aligned} \beta_{s,t} &= \beta_{s,t-1} + \eta_t, & \eta_t &\sim \mathcal{N}(0, Q), \\ r_{s,t}^e &= \beta_{s,t} r_{\text{mkt},t}^e + \varepsilon_t, & \varepsilon_t &\sim \mathcal{N}(0, R). \end{aligned}$$

We set $Q = 10^{-4}$ and estimate R from OLS residual variance. The filter can be initialized with the first return observation; nevertheless, we still enforce the global one-month lag in the sleeve assignment (month t uses $\beta_{s,t-1}$). First valid classification occurs in August 1963.

Monthly sleeve formation. For each month t , we rank the 25 signals by the lagged beta estimate $\beta_{s,t-1}$ from one of the above methods and form High- and Low-Beta sleeves (top/bottom 50%). These monthly sleeves are inputs to the annual regime allocation described in Appendix B.5 (Table 9).

TABLE 7
Complete Feature Set (35 Features)

Feature	Formula	Window
Momentum Features (6)		
momentum_3m	$\frac{1}{50} \sum_{i=1}^{50} \bar{r}_{i,[t-3:t-1]}$	3m
momentum_6m	$\frac{1}{50} \sum_{i=1}^{50} \bar{r}_{i,[t-6:t-1]}$	6m
momentum_12m	$\frac{1}{50} \sum_{i=1}^{50} \bar{r}_{i,[t-12:t-1]}$	12m
momentum_3m_median	$\text{median}(\bar{r}_{i,[t-3:t-1]})$	3m
momentum_6m_median	$\text{median}(\bar{r}_{i,[t-6:t-1]})$	6m
momentum_trend_6m	$\frac{1}{50} \sum_{i=1}^{50} \left[\prod_{k=t-6}^{t-1} (1 + r_{i,k}) - 1 \right]$	6m
Dispersion & Volatility Features (8)		
dispersion_5m	$\text{std}_i(\bar{r}_{i,[t-5:t-1]})$	5m
dispersion_12m	$\text{std}_i(\bar{r}_{i,[t-12:t-1]})$	12m
downside_vol_6m	$\frac{1}{50} \sum_{i=1}^{50} \text{std}(r_{i,j} \mid r_{i,j} < 0)_{[t-6:t-1]}$	6m
upside_vol_6m	$\frac{1}{50} \sum_{i=1}^{50} \text{std}(r_{i,j} \mid r_{i,j} > 0)_{[t-6:t-1]}$	6m
vol_asymmetry_6m	$\text{downside_vol_6m} / \text{upside_vol_6m}$	6m
max_drawdown_12m	$\frac{1}{50} \sum_{i=1}^{50} \min_j \left(\frac{P_{i,j}}{\max_{k \leq j} P_{i,k}} - 1 \right)$	12m
avg_vol_3m	$\frac{1}{50} \sum_{i=1}^{50} \sigma_{i,[t-3:t-1]}$	3m
vol_of_vol_6m	$\frac{1}{50} \sum_{i=1}^{50} \text{std}(\sigma_{i,[t-6:t-1]})$	6m
Beta Features (7)		
beta_range	$\max_j \beta_{j,t-1} - \min_j \beta_{j,t-1}$	PIT
beta_dispersion	$\text{std}_j(\beta_{j,t-1})$	PIT
beta_skewness	$\text{skew}_j(\beta_{j,t-1})$	PIT
beta_mean	$\frac{1}{25} \sum_{j=1}^{25} \beta_{j,t-1}$	PIT
beta_momentum_3m	$\frac{1}{25} \sum_{j=1}^{25} (\beta_{j,t-1} - \beta_{j,t-4})$	3m
beta_momentum_6m	$\frac{1}{25} \sum_{j=1}^{25} (\beta_{j,t-1} - \beta_{j,t-7})$	6m
beta_trend_12m	$\frac{1}{25} \sum_{j=1}^{25} \text{slope}(\beta_{j,[t-12:t-1]})$	12m
Cross-Sectional Features (8)		
pct_positive	$\frac{1}{50} \sum_{i=1}^{50} \mathbb{1}(r_{i,t-1} > 0)$	1m
pct_positive_3m	$\frac{1}{50} \sum_{i=1}^{50} \mathbb{1}(\bar{r}_{i,[t-3:t-1]} > 0)$	3m
concentration_hhi	$\sum_{i=1}^{50} \left(\frac{ r_{i,t-1} }{\sum_j r_{j,t-1} } \right)^2$	1m
skewness_cs	$\text{skew}_i(r_{i,t-1})$	1m
kurtosis_cs	$\text{kurt}_i(r_{i,t-1})$	1m
tail_ratio	$(P_{90} - P_{50}) / (P_{50} - P_{10})$	1m
top_bottom_spread	$P_{90}(r_{t-1}) - P_{10}(r_{t-1})$	1m
median_return	$\text{median}_i(r_{i,t-1})$	1m
Interaction Features (6)		
mom_x_disp	$\text{momentum_6m} \times \text{dispersion_5m}$	Comp.
mom_x_beta_range	$\text{momentum_6m} \times \text{beta_range}$	Comp.
beta_range_x_disp	$\text{beta_range} \times \text{dispersion_5m}$	Comp.
downside_x_beta	$\text{downside_vol_6m} \times \text{beta_dispersion}$	Comp.
momentum_sharpe	$\text{momentum_6m} / (\text{dispersion_5m} + 10^{-6})$	Comp.
beta_vol_ratio	$\text{beta_dispersion} / (\text{avg_vol_3m} + 10^{-6})$	Comp.

TABLE 8
Hyperparameter Specifications Across All Models

Model	Parameter	Value
Logistic Regression	Penalty	L2 (Ridge)
	Regularization strength (C)	1.0
	Solver	LBFGS
	Max iterations	1000
	Class weight	None
	Random state	42
Random Forest	Number of trees	200
	Max depth	3
	Max features	$\log_2(35) \approx 5$
	Min samples leaf	5
	Class weight	None
	Random state	42
XGBoost	Number of estimators	200
	Max depth	5
	Learning rate (η)	0.1
	Subsample	0.8
	Colsample by tree	0.7
	L1 regularization (α)	0.0
	L2 regularization (λ)	0.1
	Min child weight	1
	Gamma	0.0
	Scale pos weight	$n_{\text{LOW}}/n_{\text{HIGH}} \approx 1.70$
	Tree method	Histogram
	Objective	Binary logistic
	Random state	42
LSTM	Architecture	LSTM(32) + Dropout(0.2) + Dense(1, sigmoid)
	LSTM units	32
	Dropout rate	0.2
	Optimizer	Adam ($\alpha = 0.001$)
	Loss function	Binary crossentropy
	Batch size	32
	Epochs	30
	Early stopping	Patience = 3 (monitor: loss)
	Random state	42
	Window (<i>Expanding</i>)	Variable (min 60 months, max 648 months)
	Window (<i>Rolling 10Y</i>)	Fixed (120 months)

Notes: Hyperparameter specifications for all machine learning models. Logistic Regression serves as the linear baseline with L2 regularization. Random Forest uses shallow trees (depth=3) with $\log_2$ feature sampling to reduce overfitting. XGBoost employs gradient boosting with histogram-based splits and class-weighted loss (≈ 1.70) to handle imbalanced regimes. LSTM uses a single-layer recurrent architecture with two training strategies: Expanding (variable window, 60–648 months) and Rolling 10-Year (fixed 120-month window). All models use random state 42 for reproducibility.

TABLE 9
Comprehensive Timing Protocol Summary

Component	Frequency	Formation Date	Information Set	Warmup / Buffer / First Valid
Stage 1: Signal Portfolios				
P10/P1 portfolios	Annual	June 30, y	FY($y - 1$), ME(May y)	Jun 1963 (first formation)
Holding period	Monthly	Jul(y) – Jun($y + 1$)	–	12-month rebalancing cycle
Stage 2: Basket Construction				
Basket 1 (Fixed)	Annual	June 30, y	Economic classification	Jun 1963 (no warmup)
Baskets 2-5 (Beta)	Monthly	End of month $t - 1$	$\beta_{s,t-1}$ (lagged)	Warmup: 60m $\rightarrow$ Jul 1968
Stage 3: Features & Label				
Feature matrix X_t	Monthly	End of month $t - 1$	Returns through $t - 1$	Warmup: 60m $\rightarrow$ Jun 1969
Regime label Y_t	Monthly	End of month $t - 1$	EWMA vol through $t - 1$	Warmup: 60m $\rightarrow$ Jun 1969
Stage 4: ML Training & Backtest				
Model refit	Annual	June 30, y	Train: Jun 1969 – Jun($y - 1$)	Warmup: 121m (10Y) $\rightarrow$ Jun 1980 Buffer: Jun 1969 – Jun 1980
Forecast horizon	12 months	Jul(y) – Jun($y + 1$)	Single prediction per year	Annual forecast
Out-of-sample test	Monthly	Jul 1980 – Jun 2024	528 months, 44 years	OOS period

Note: This table documents timing conventions across all pipeline stages. **Warmup periods** ensure stable estimation: 60 months for beta/EWMA/feature initialization (first valid: Jul 1968 for betas, Jun 1969 for features/label), and 121 months for ML training (first valid: Jun 1980 requiring 10-year history). **Buffer period** (Jun 1969 – Jun 1980) separates warmup completion from out-of-sample testing, ensuring all components are fully aligned. All formation decisions at time t use information sets $\mathcal{F}_{t-1}$, enforced via systematic 1-month lags on betas, features, and regime labels. The out-of-sample period (Jul 1980 – Jun 2024) is identical across all models, ensuring fair comparison.

385 B.5. Timing protocol.

TABLE 10
Model Training Protocols and Comparability

Model	Strategy	First Train Start	First Train End	First Size	Last Size	Formations	OOS Period
Logistic Reg.	Expanding	1969-06	1979-06	121 months	637 months	44	Jul 1980 – Jun 2024
Random Forest	Expanding	1969-06	1979-06	121 months	637 months	44	Jul 1980 – Jun 2024
XGBoost	Expanding	1969-06	1979-06	121 months	637 months	44	Jul 1980 – Jun 2024
LSTM-Expanding	Expanding	1969-06	1979-06	121 months	637 months	44	Jul 1980 – Jun 2024
LSTM-Rolling 10Y	Rolling	1970-07	1979-06	120 months	120 months	44	Jul 1980 – Jun 2024

Note: All models share identical out-of-sample testing periods (Jul 1980 – Jun 2024, 528 months, 44 years), ensuring fair performance comparison. Logistic, Random Forest, XGBoost, and LSTM-Expanding employ expanding window protocols with first training spanning Jun 1969 – Jun 1979 (121 months). LSTM-Rolling 10Y uses a fixed 120-month window by design, starting one month later (Jul 1970) to maintain exactly 10 years. This minor 1-month offset (<1% difference in training data) does not affect comparability. All models produce 44 annual formations (1980–2023), with each formation generating predictions for the subsequent 12-month holding period.

386 B.6. Model comparability.

Appendix C. Data.

TABLE 11
Sample construction and filtering procedure

Period	Years	Raw CRSP	After Exchange	After NYSE size filter	Eligible $\geq \$5$
Early Period	1963–1979	3,045	2,975	1,167	1,152
Middle Period	1980–1999	6,799	6,645	2,435	2,391
Recent Period	2000–2024	6,088	5,960	3,278	3,128
Overall	1963–2024	5,483	5,363	2,427	2,348

Notes: This table reports the average number of firms at each stage of the sample construction across three subperiods. *Raw CRSP* includes all securities in CRSP. *After Exchange* restricts to NYSE/AMEX/NASDAQ. *After NYSE size filter* applies the “all-but-microcaps” screen using the NYSE 20th-percentile market-cap breakpoint (computed from NYSE firms and applied to the full universe). *Eligible $\geq \$5$* further imposes the price screen at the June formation date. Sample period: July 1963 to June 2024.

C.1. Sample construction and filtering procedure.

C.2. Delisting Treatment. A well-documented challenge in long-horizon return studies is the treatment of delisted securities. (14) CRSP provides delisting returns (DLRET) for most delistings, but approximately 30% of performance-related delistings (mainly liquidations and non-compliance removals) have missing delisting returns. Shumway (1997) (14) shows that ignoring these missing delisting returns induces a substantial upward bias in average returns, because delistings for poor performance are typically associated with large negative final returns.

Motivated by this evidence, we correct missing performance-related delisting returns using a stochastic imputation scheme designed to mimic the distribution of observed delisting returns:

- **Bankruptcy Delistings** (DLSTCD 400–490):

To account for exchange-specific recovery rates consistent with Shumway (1997), we differentiate by listing venue:

- *NYSE/AMEX*: Impute a delisting return drawn from $\mathcal{N}(-30\%, 15\%)$, reflecting higher average recovery rates.
- *NASDAQ*: Impute a delisting return drawn from $\mathcal{N}(-55\%, 15\%)$, reflecting more severe losses.

All draws are truncated to $[-95\%, -5\%]$ to rule out implausible values.

- **Non-Compliance Delistings** (DLSTCD 500–590):

Impute a delisting return drawn from $\mathcal{N}(-30\%, 10\%)$, truncated to $[-60\%, -5\%]$, regardless of the exchange.

The means and truncation ranges are chosen to produce loss magnitudes consistent with the severe underperformance documented for performance-related delistings (14), while ruling out implausible values. To ensure reproducibility of the stochastic imputation, we employ firm-specific random seeds based on permanent identifiers (PERMNO). The adjusted return for delisting month t is then computed as:

$$r_{i,t}^{\text{adj}} = (1 + r_{i,t}) \times (1 + \text{DLRET}_{i,t}) - 1,$$

where $r_{i,t}$ is the regular monthly return and $\text{DLRET}_{i,t}$ is either the observed or imputed delisting return. Following imputation, we apply a “zombie filter” to remove all post-delisting observations from the panel. This ensures that firms exit the sample

419 permanently upon fatal delisting, consistent with the economic reality of bankruptcy
 420 or exchange removal.

TABLE 12
Portfolio Statistics Summary by Signal (1963–2024)

Category	Signal	Total	Avg/Port	Min	Max	Median	Small%	Big%	Low%	High%
<i>Panel A: Profitability</i>										
	Gross Profitability	1,889	75.6	48	135	65	29.0	14.9	22.3	22.9
	Operating Profitability	1,541	61.7	32	151	58	29.3	14.5	26.1	20.9
	Return on Assets	1,903	76.1	40	176	72	29.0	14.9	24.0	22.3
	Δ Asset Turnover	1,808	72.3	45	126	67	28.5	15.2	20.8	20.9
	Δ Profit Margin	1,790	71.6	46	121	65	28.4	15.2	20.7	21.8
<i>Panel B: Investment</i>										
	Return on Invested Capital	2,118	84.7	39	165	78	29.4	14.7	20.8	23.6
	Asset Growth	1,815	72.6	41	141	66	28.6	15.2	22.1	25.1
	Investment (CapEx)	1,815	72.6	41	141	66	28.6	15.2	22.1	25.1
	CapEx + Inventory Growth	1,555	62.2	37	104	57	28.4	15.5	22.6	22.1
	Net Share Issuance	2,235	89.4	47	142	90	27.6	15.0	23.5	23.5
	CapEx Growth	1,655	66.2	39	113	59	28.3	15.5	20.8	22.6
<i>Panel C: Accruals & Quality</i>										
	Sales Growth	1,801	72.1	48	146	62	28.4	15.2	20.9	25.2
	Working Capital Accruals	1,433	57.3	32	119	55	28.4	15.5	21.7	23.3
	Sales-Receivables Gap	1,745	69.8	47	134	61	28.6	15.1	22.1	23.4
	Receivables Turnover	1,819	72.8	50	147	66	28.9	14.9	24.0	18.9
<i>Panel D: Liquidity & Solvency</i>										
	Operating Leverage	1,728	69.1	48	135	62	28.7	15.2	21.0	24.1
	Cash Cushion	1,882	75.3	47	192	64	29.1	14.9	16.4	30.1
	Book Leverage	2,109	84.4	55	220	72	29.4	14.7	29.0	18.4
	Δ Current Ratio	1,462	58.5	36	114	53	28.2	15.6	23.0	23.2
	Debt Capacity	1,748	69.9	39	148	65	29.6	14.7	26.8	17.2
<i>Panel E: Operating Efficiency</i>										
	Cash Productivity	1,726	69.0	41	170	58	28.7	15.1	29.5	16.7
	Depreciation/PPE	1,712	68.5	44	155	59	29.4	15.0	16.7	25.8
	Δ Depreciation	1,686	67.5	36	123	60	28.9	15.2	23.5	23.6
	Tax/Income Ratio	1,907	76.3	41	153	70	28.9	14.9	22.7	19.8
	Asset Turnover	1,907	76.3	39	133	69	28.9	14.9	22.4	19.6
MEAN		1,791	71.7	42	144	64	28.8	15.1	22.6	22.4

Notes: This table reports time-series averages of portfolio characteristics across 25 fundamental signals. Each signal is independently sorted into quintiles based on NYSE breakpoints. The sample includes U.S. common stocks (1963–2024) that pass the Fama-French NYSE size filter (20th percentile and price $\geq \$5$) screen. *Total* is the average number of firms across all 25 portfolios (5 size $\times$ 5 signal quintiles). *Avg/Port* is the mean portfolio size. *Min/Max* indicate the smallest and largest portfolios. *Small%/Big%* report the percentage of firms in the small-cap and large-cap quintiles. *Low%/High%* report the percentage of firms in the lowest and highest signal quintiles. Signals are grouped into five categories following the factor literature: Profitability (Novy-Marx, 2013), Investment (Cooper et al., 2008), Accruals & Quality (Sloan, 1996; Richardson et al., 2005), Liquidity & Solvency (Piotroski, 2000), and Operating Efficiency (Soliman, 2008).

421 C.3. Size $\times$ Signal portfolio statistics.

Appendix D. Results.

D.1. . Full Performance Grid (Gross vs. Net)

TABLE 13
Complete Performance Metrics: Gross vs. Net Returns (1980–2024)

Gross Returns (Signal Power)								Net Returns (Realizability)								
Basket	Model	Ann.Ret. (%)	CAGR (%)	Sharpe	Sortino	Max DD (%)	Date	Basket	Model	Ann.Ret. (%)	CAGR (%)	Sharpe	Sortino	Max DD (%)	Date	
Economic	XGB	15.93	14.78	0.596	0.860	-49.0	2009-02	XGB	15.54	14.34	0.577	0.832	-49.3	2009-02		
	RF	15.56	14.35	0.577	0.841	-49.0	2009-02	RF	15.17	13.91	0.557	0.812	-49.3	2009-02		
	Logistic L2	15.49	14.29	0.575	0.834	-49.0	2009-02	Logistic L2	15.10	13.85	0.555	0.806	-49.3	2009-02		
	LSTM Rolling 10Y	15.30	14.09	0.568	0.832	-49.0	2009-02	LSTM Rolling 10Y	14.91	13.85	0.548	0.804	-49.3	2009-02		
	LSTM Expanding	15.32	14.10	0.566	0.827	-49.0	2009-02	LSTM Expanding	14.93	13.65	0.547	0.798	-49.3	2009-02		
Beta Exp.	LSTM Rolling 10Y	16.85	15.88	0.649	0.933	-47.5	2009-02	LSTM Rolling 10Y	16.47	15.45	0.631	0.906	-47.8	2009-02		
	LSTM Expanding	16.76	15.71	0.635	0.916	-47.5	2009-02	LSTM Expanding	16.38	15.28	0.617	0.889	-47.8	2009-02		
	RF	16.65	15.57	0.627	0.916	-47.5	2009-02	RF	16.28	15.14	0.608	0.880	-47.8	2009-02		
	Logistic L2	16.39	15.29	0.617	0.885	-47.5	2009-02	Logistic L2	16.01	14.86	0.599	0.858	-47.8	2009-02		
	XGB	16.05	14.84	0.591	0.877	-47.5	2009-02	XGB	15.68	14.42	0.573	0.849	-47.8	2009-02		
Rolling β	LSTM Rolling 10Y	16.71	15.75	0.646	0.931	-44.8	2009-02	LSTM Rolling 10Y	16.34	15.32	0.627	0.903	-45.1	2009-02		
	LSTM Expanding	16.55	15.49	0.628	0.906	-44.8	2009-02	LSTM Expanding	16.17	15.06	0.609	0.878	-45.1	2009-02		
	RF	16.57	15.44	0.617	0.908	-44.8	2009-02	RF	16.19	15.01	0.599	0.880	-45.1	2009-02		
	Logistic L2	16.20	15.08	0.609	0.875	-44.8	2009-02	Logistic L2	15.82	14.65	0.590	0.848	-45.1	2009-02		
	XGB	15.87	14.61	0.578	0.864	-44.8	2009-02	XGB	15.49	14.18	0.560	0.835	-45.1	2009-02		
EWMA β	LSTM Rolling 10Y	16.55	15.57	0.638	0.913	-45.8	2009-02	LSTM Rolling 10Y	16.16	15.13	0.618	0.885	-46.1	2009-02		
	LSTM Expanding	16.50	15.43	0.624	0.896	-45.8	2009-02	LSTM Expanding	16.12	14.99	0.604	0.869	-46.1	2009-02		
	Logistic L2	16.31	15.19	0.612	0.880	-45.8	2009-02	Logistic L2	15.93	14.75	0.593	0.853	-46.1	2009-02		
	RF	16.35	15.17	0.604	0.876	-45.8	2009-02	RF	15.96	14.73	0.586	0.850	-46.1	2009-02		
	XGB	15.67	14.36	0.566	0.840	-45.8	2009-02	XGB	15.29	13.92	0.547	0.812	-46.1	2009-02		
Kahn β	LSTM Rolling 10Y	16.58	15.60	0.639	0.918	-44.9	2009-02	LSTM Rolling 10Y	16.20	15.16	0.620	0.890	-45.2	2009-02		
	LSTM Expanding	16.62	15.56	0.629	0.909	-44.9	2009-02	LSTM Expanding	16.24	15.12	0.610	0.881	-45.2	2009-02		
	RF	16.50	15.35	0.613	0.897	-44.9	2009-02	RF	16.11	14.91	0.594	0.869	-45.2	2009-02		
	Logistic L2	16.26	15.13	0.609	0.878	-44.9	2009-02	Logistic L2	15.87	14.69	0.590	0.850	-45.2	2009-02		
	XGB	15.81	14.54	0.575	0.858	-44.9	2009-02	XGB	15.43	14.11	0.557	0.829	-45.2	2009-02		
NYSE Benchmark		14.71	13.95	0.613	0.846	-48.4	2009-02	NYSE Benchmark		14.71	13.95	0.613	0.846	-48.4	2009-02	

Notes: This table presents the complete performance metrics for all 25 basket-model combinations under gross returns (left panel, signal power) and net returns (right panel, realizability after transaction costs). The sample consists of NYSE/AMEX/NASDAQ stocks with price $\geq \$5$ and market capitalization ≥ 20 th percentile (NYSE) from July 1980 to June 2024 (528 months). Within each basket, models are sorted by Sharpe ratio (descending). Net returns incorporate Frazzini (2018) transaction costs applied annually in June. The NYSE Benchmark is an equal-weighted portfolio of all eligible stocks with monthly rebalancing (no transaction costs, identical for both panels). $CAGR$ = compound annual growth rate, $Sharpe$ = Sharpe ratio, $Sortino$ = Sortino ratio, $Max DD$ = maximum drawdown, $Date$ = date of maximum drawdown. Requires packages `pdfscape` and `booktabs`.

TABLE 14
Basket-Level Performance Analysis: Offensive vs Defensive Strategies (1980–2024)

Panel	Basket	Regime	Ann. Ret. (%)	CAGR (%)	Vol. (%)	Sharpe	Sortino	Max DD (%)	DD Date
Panel A: Gross Returns (No Transaction Costs)									
	Economic Basket	Offensive	15.64	14.41	20.36	0.577	0.840	-49.0	2009-02
		Defensive	15.67	14.62	19.68	0.599	0.873	-46.6	2009-02
	Beta Expansion	Offensive	16.08	14.82	20.80	0.587	0.869	-47.5	2009-02
		Defensive	16.11	15.35	18.62	0.657	0.929	-47.0	2009-02
	Rolling Beta	Offensive	16.25	14.94	21.12	0.585	0.877	-44.8	2009-02
		Defensive	15.86	15.10	18.39	0.652	0.912	-49.2	2009-02
	EWMA Beta	Offensive	16.06	14.70	21.22	0.574	0.857	-45.8	2009-02
		Defensive	16.04	15.32	18.32	0.664	0.924	-48.0	2009-02
	Kalman Filter Beta	Offensive	16.20	14.89	21.14	0.583	0.876	-44.9	2009-02
		Defensive	15.85	15.11	18.32	0.654	0.911	-48.0	2009-02
	NYSE Benchmark	—	14.71	13.95	17.66	0.613	0.846	-48.4	2009-02
Panel B: Net Returns (With Transaction Costs)									
	Economic Basket	Offensive	15.24	13.97	20.37	0.558	0.812	-49.3	2009-02
		Defensive	15.33	14.22	19.69	0.581	0.847	-46.8	2009-02
	Beta Expansion	Offensive	15.69	14.38	20.81	0.567	0.840	-47.7	2009-02
		Defensive	15.75	14.94	18.62	0.638	0.903	-47.4	2009-02
	Rolling Beta	Offensive	15.86	14.50	21.13	0.566	0.847	-45.0	2009-02
		Defensive	15.49	14.69	18.39	0.632	0.885	-49.4	2009-02
	EWMA Beta	Offensive	15.65	14.23	21.23	0.554	0.826	-46.1	2009-02
		Defensive	15.68	14.91	18.32	0.645	0.897	-48.3	2009-02
	Kalman Filter Beta	Offensive	15.81	14.44	21.15	0.564	0.846	-45.1	2009-02
		Defensive	15.48	14.69	18.32	0.633	0.885	-48.4	2009-02
	NYSE Benchmark	—	14.71	13.95	17.66	0.613	0.846	-48.4	2009-02

Notes: Performance metrics averaged across all ML models within each (Basket, Regime) pair. Transaction costs applied once per year in June following Fama-French convention. Inter-basket costs (High/Low Beta switching) included in strategy-specific TC. Baskets sorted by Sharpe ratio (descending) within each panel.

D.2. Basket Building Blocks: Offensive vs. Defensive Sleeves (Gross vs. Net).

TABLE 15
Portfolio Turnover by Basket Strategy (1980–2024)

Basket Strategy	Panel A: Aggregate			Panel B: By Period								
	Stock	Inter	Total	1980–1999			2000–2009			2010–2024		
	(%)	(%)	(%)	Stock	Inter	Total	Stock	Inter	Total	Stock	Inter	Total
Economic Basket	100.9	0.0	100.9	116.7	0.0	116.7	106.3	0.0	106.3	77.3	0.0	77.3
Beta Expansion	115.9	0.0	115.9	138.1	0.0	138.1	117.3	0.0	117.3	86.8	0.0	86.8
Rolling Beta	124.1	2.5	126.6	137.2	1.3	138.5	143.2	4.0	147.2	94.7	3.2	97.9
EWMA Beta	120.9	5.3	126.2	131.4	5.9	137.3	137.7	4.8	142.5	96.6	4.8	101.4
Kalman Filter Beta	120.9	3.3	124.2	133.1	2.1	135.2	138.8	3.2	142.0	93.6	4.8	98.4

Note: This table reports portfolio turnover for different basket strategies from 1980 to 2024. Panel A shows the aggregate (time-averaged) turnover, while Panel B breaks down turnover by three periods: 1980–1999, 2000–2009, and 2010–2024. Turnover is measured as the annual percentage of the portfolio that is rebalanced. “Stock” refers to stock-level turnover from annual June rebalancing (Fama-French convention). “Inter” refers to inter-basket turnover from switches between Offensive/Defensive or High Beta/Low Beta strategies. All values are in percentage points (%).

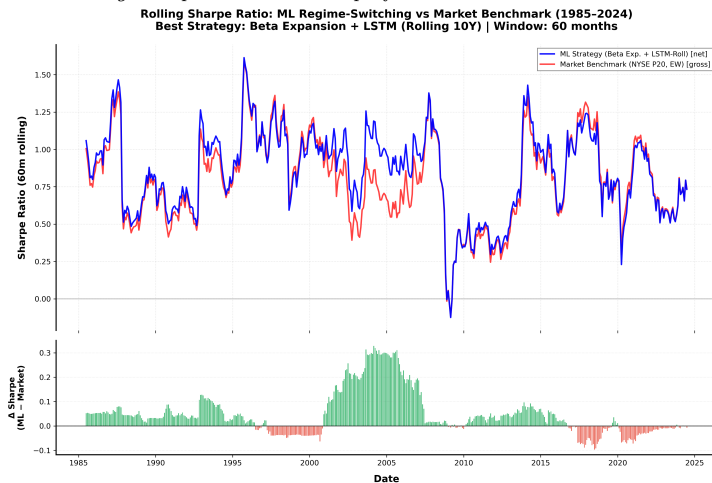
424 D.3. Turnover and Transaction Costs.

TABLE 16
Transaction Costs by Basket Strategy (1980–2024)

Basket Strategy	Panel A: Aggregate			Panel B: By Period								
	Stock	Inter	Total	1980–1999			2000–2009			2010–2024		
	(bps)	(bps)	(bps)	Stock	Inter	Total	Stock	Inter	Total	Stock	Inter	Total
Economic	37.0	0.0	37.0	60.7	0.0	60.7	23.7	0.0	23.7	15.8	0.0	15.8
Beta Expansion	37.5	0.0	37.5	62.6	0.0	62.6	23.3	0.0	23.3	15.3	0.0	15.3
Rolling Beta	37.1	0.7	37.7	61.4	0.6	62.0	23.4	0.8	24.2	15.3	0.6	16.0
EWMA Beta	36.6	1.8	38.4	60.4	2.9	63.4	23.0	1.0	24.0	15.3	1.0	16.3
Kalman Beta	37.2	0.9	38.1	62.0	1.1	63.0	23.2	0.6	23.8	15.1	1.0	16.0
<i>Mean</i>	<i>39.5</i>	<i>0.7</i>	<i>40.2</i>	<i>60.8</i>	<i>0.9</i>	<i>61.7</i>	<i>35.4</i>	<i>0.4</i>	<i>35.7</i>	<i>15.4</i>	<i>0.6</i>	<i>16.0</i>

Notes: This table reports annual transaction costs for each basket strategy. *Panel A* presents aggregate costs (July 1980–June 2024, 528 months). *Panel B* decomposes costs by period. *Stock* TC reflects one-way costs at annual rebalancing (June) following Frazzini, Israel, and Moskowitz (2012): 50 bps (1980–1999), 30 bps (2000–2009), 20 bps (2010–2024). *Inter* TC captures switching costs between High/Low Beta portfolios (Baskets 2–5 only; Basket 1 uses Economic cycles, hence Inter = 0). *Total* = Stock + Inter. All costs are annualized and expressed in basis points. The NYSE Benchmark is reported gross (no transaction costs applied).

425 D.3.1. Transaction Costs by Basket Strategy.

FIG. 4. *Rolling Sharpe Ratio and Outperformance: ML vs Market Benchmark*

Notes: *Panel A* shows the 60-month rolling Sharpe ratio (annualized) for the best-performing ML regime-switching strategy (solid blue line, net of transaction costs) and the market benchmark (red line, NYSE 20th-percentile market-cap breakpoint equal-weighted portfolio, gross returns). *Panel B* displays the difference in Sharpe ratios ($\Delta \text{Sharpe} = \text{ML} - \text{Market}$), with green bars indicating periods where the ML strategy outperforms (positive Δ) and red bars indicating underperformance (negative Δ). The ML strategy is selected based on the maximum net Sharpe ratio over the full sample (July 1980–June 2024). Rolling Sharpe ratios are computed using a 60-month backward-looking window, with the first observation in January 1985. The ML strategy outperforms the market over extended subperiods, although episodes of underperformance also occur. The benchmark is shown gross (no transaction costs), implying that ΔSharpe is conservative with respect to net implementability.

D.4. Time-Varying Performance: Rolling Sharpe and Relative Outperformance.

TABLE 17
Comprehensive Factor Analysis: Classic Baskets and ML Strategies (Net Returns)

Basket	Strategy	FF5			FF6			FF7			Q5				
		α	t	R^2	α	t	R^2	α	t	R^2	α	t	R^2		
Panel A: Classic Buy&Hold Baskets (Net)															
Economic	(Offensive)	-0.29	-0.23	0.924	1.26		0.96	0.939	1.13	0.79	0.939	2.11	1.31	0.904	
	(Defensive)	0.90	0.77	0.943	2.14	*	1.68	0.954	1.95	1.42	0.954	2.22	1.36	0.936	
Beta Exp.	(Offensive)	1.39	1.04	0.938	2.82	*	1.94	0.951	2.60	*	1.66	0.951	3.00	1.59	0.934
	(Defensive)	0.01	0.01	0.933	1.28		1.17	0.945	1.15	0.99	0.945	1.80	1.22	0.911	
Rolling β	(Offensive)	1.39	0.96	0.932	2.91	*	1.80	0.945	2.70	1.57	0.945	3.25	1.57	0.929	
	(Defensive)	-0.11	-0.10	0.929	1.08		1.03	0.939	0.94	0.83	0.940	1.37	0.96	0.903	
EWMA β	(Offensive)	1.18	0.84	0.933	2.62	*	1.70	0.945	2.38	1.45	0.946	2.98	1.51	0.932	
	(Defensive)	0.16	0.14	0.927	1.33		1.19	0.937	1.21	0.99	0.938	1.64	1.12	0.900	
Kalman β	(Offensive)	1.33	0.95	0.933	2.78	*	1.84	0.945	2.49	1.54	0.946	3.03	1.56	0.931	
	(Defensive)	-0.14	-0.12	0.927	1.08		0.96	0.938	1.00	0.82	0.939	1.49	0.99	0.901	
Panel B: Machine Learning Strategies (Net)															
Economic	Logistic	0.32	0.26	0.933	1.66		1.27	0.945	1.40	0.99	0.945	2.10	1.31	0.923	
	RF	-0.28	-0.23	0.926	1.25		0.98	0.941	1.06	0.77	0.942	1.90	1.21	0.909	
	XGB	0.25	0.20	0.927	1.73		1.31	0.941	1.54	1.08	0.941	2.14	1.32	0.908	
	LSTM-Exp	-0.08	-0.06	0.925	1.42		1.14	0.940	1.17	0.88	0.941	2.03	1.27	0.913	
	LSTM-Roll	-0.17	-0.14	0.927	1.31		1.06	0.941	1.09	0.82	0.942	1.90	1.19	0.915	
Beta Exp.	Logistic	0.37	0.31	0.931	1.86	*	1.66	0.946	1.56	1.26	0.946	1.84	1.25	0.916	
	RF	1.49	1.12	0.938	2.77	*	1.94	0.948	2.53	*	1.66	0.949	2.77	1.49	0.932
	XGB	1.10	0.78	0.935	2.53	*	1.66	0.947	2.26	1.36	0.948	2.67	1.32	0.930	
	LSTM-Exp	0.90	0.83	0.947	2.00	*	1.80	0.955	1.76	1.46	0.956	1.95	1.55	0.936	
	LSTM-Roll	1.12	1.03	0.947	2.16	*	1.92	0.954	2.06	*	1.71	0.954	2.28	*	1.74
Rolling β	Logistic	0.23	0.18	0.918	1.77		1.52	0.934	1.45	1.12	0.934	1.42	0.93	0.902	
	RF	1.80	1.26	0.929	3.14	**	2.02	0.940	2.85	*	1.71	0.941	2.99	1.49	0.926
	XGB	1.30	0.87	0.926	2.85	*	1.74	0.941	2.55	1.44	0.942	2.84	1.30	0.924	
	LSTM-Exp	1.16	0.98	0.941	2.21	*	1.83	0.948	1.94	1.48	0.948	1.75	1.39	0.933	
	LSTM-Roll	1.35	1.14	0.940	2.39	**	1.99	0.947	2.24	*	1.71	0.948	2.09	1.59	0.931
EWMA β	Logistic	0.28	0.21	0.915	1.84		1.54	0.931	1.49	1.12	0.932	1.36	0.87	0.899	
	RF	1.56	1.09	0.929	2.80	*	1.83	0.938	2.47	1.51	0.939	2.58	1.31	0.927	
	XGB	1.06	0.72	0.926	2.54		1.60	0.940	2.17	1.26	0.941	2.45	1.15	0.925	
	LSTM-Exp	1.12	0.88	0.937	2.17	*	1.65	0.944	1.86	1.30	0.945	1.52	1.15	0.930	
	LSTM-Roll	1.23	0.98	0.936	2.29	*	1.76	0.944	2.15	1.52	0.944	1.81	1.32	0.928	
Kalman β	Logistic	0.33	0.25	0.919	1.82		1.51	0.933	1.49	1.11	0.934	1.47	0.94	0.903	
	RF	1.56	1.12	0.930	2.82	*	1.91	0.940	2.46	1.55	0.941	2.59	1.35	0.927	
	XGB	1.05	0.72	0.927	2.52		1.61	0.940	2.12	1.25	0.942	2.40	1.14	0.925	
	LSTM-Exp	1.18	0.98	0.939	2.23	*	1.81	0.946	1.91	1.41	0.947	1.73	1.32	0.931	
	LSTM-Roll	1.14	0.94	0.938	2.17	*	1.73	0.945	2.02	1.48	0.946	1.86	1.35	0.929	

Notes: This table reports risk-adjusted alphas (α , in % per annum), t -statistics (Newey-West HAC, 12 lags), and R^2 for all ML strategies across five baskets and four factor models. All returns are net of transaction costs. FF5/FF6/FF7 are Fama–French factor models: FF5 includes market (MKT), size (SMB), value (HML), profitability (RMW), and investment (CMA); FF6 extends FF5 by adding momentum (UMD); and **FF7 extends FF6 by adding quality-minus-junk (QMJ)**. Q5 refers to the Hou–Xue–Zhang q -factor model (market, size, investment, profitability, and expected growth). Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Sample: July 1980–June 2024.

D.5. Comprehensive Factor Regressions.

D.6. Incremental Alpha Tests.

TABLE 18
Incremental Alpha: All ML Strategies vs 50/50 Benchmark (Full)

Basket	Model	FF6				Q5			
		$\Delta\alpha$	t	p	q (FDR)	$\Delta\alpha$	t	p	q (FDR)
Economic	Logistic	-0.04	-0.14	0.890	0.927	-0.06	-0.21	0.830	0.899
	RF	-0.45	-1.54	0.124	0.636	-0.26	-0.83	0.408	0.899
	XGB	0.03	0.12	0.907	0.927	-0.03	-0.10	0.918	0.918
	LSTM-Exp	-0.28	-0.88	0.378	0.859	-0.13	-0.46	0.649	0.899
	LSTM-Roll	-0.39	-1.23	0.219	0.781	-0.26	-0.88	0.377	0.899
Beta Exp.	Logistic	-0.19	-0.36	0.721	0.927	-0.56	-0.83	0.406	0.899
	RF	0.72	1.58	0.114	0.636	0.37	0.71	0.477	0.899
	XGB	0.48	1.04	0.300	0.825	0.27	0.49	0.624	0.899
	LSTM-Exp	-0.05	-0.09	0.927	0.927	-0.45	-0.65	0.517	0.899
	LSTM-Roll	0.10	0.20	0.844	0.927	-0.12	-0.17	0.863	0.899
Rolling β	Logistic	-0.23	-0.33	0.740	0.927	-0.89	-1.00	0.319	0.899
	RF	1.14	*	1.92	0.054	0.68	0.90	0.370	0.899
	XGB	0.86	1.44	0.151	0.636	0.53	0.66	0.508	0.899
	LSTM-Exp	0.21	0.30	0.761	0.927	-0.56	-0.62	0.534	0.899
	LSTM-Roll	0.40	0.58	0.564	0.927	-0.22	-0.25	0.803	0.899
EWMA β	Logistic	-0.13	-0.21	0.836	0.927	-0.95	-1.13	0.257	0.899
	RF	0.82	1.43	0.153	0.636	0.27	0.38	0.705	0.899
	XGB	0.57	0.97	0.330	0.825	0.14	0.18	0.860	0.899
	LSTM-Exp	0.20	0.29	0.772	0.927	-0.79	-0.95	0.341	0.899
	LSTM-Roll	0.32	0.48	0.632	0.927	-0.50	-0.64	0.520	0.899
Kalman β	Logistic	-0.11	-0.18	0.854	0.927	-0.79	-0.97	0.330	0.899
	RF	0.89	1.61	0.107	0.636	0.33	0.46	0.642	0.899
	XGB	0.59	1.04	0.297	0.825	0.14	0.19	0.852	0.899
	LSTM-Exp	0.29	0.45	0.651	0.927	-0.53	-0.64	0.521	0.899
	LSTM-Roll	0.24	0.37	0.710	0.927	-0.40	-0.51	0.611	0.899

Notes: This table reports incremental alphas ($\Delta\alpha$, annualized in % p.a.) for all 25 ML strategies (5 baskets $\times$ 5 models) against a 50/50 benchmark (50% Buy&Hold Offensive + 50% Defensive). Incremental alphas are computed by regressing the difference between ML and benchmark returns on Fama–French 6-factor (FF6) and Hou–Xue–Zhang q -factor (Q5) models. t -statistics use Newey–West HAC standard errors (12 lags) on monthly data. p = raw p -value; q (FDR) = Benjamini–Hochberg FDR-adjusted p -value ($\alpha = 10\%$). All returns are net of transaction costs. Sample: July 1980–June 2024. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

D.6.1. All ML Strategies vs 50/50 Benchmark.

TABLE 19
Incremental Alpha Robustness: ML vs Offensive and Defensive Benchmarks (FF6)

Basket	Model	vs Offensive				vs Defensive			
		$\Delta\alpha$		t	p	$\Delta\alpha$		t	p
Economic	Logistic	0.40	*	1.65	0.099	-0.47		-1.07	0.284
	RF	-0.01		-0.06	0.949	-0.89	*	-1.83	0.067
	XGB	0.46	*	1.71	0.087	-0.41		-1.06	0.290
	LSTM-Exp	0.15		0.55	0.580	-0.72		-1.48	0.138
	LSTM-Roll	0.04		0.14	0.887	-0.83	*	-1.80	0.072
Beta Exp.	Logistic	-0.96		-1.41	0.159	0.58		0.93	0.351
	RF	-0.05		-0.13	0.895	1.49	**	2.05	0.041
	XGB	-0.29		-0.71	0.476	1.25	*	1.69	0.091
	LSTM-Exp	-0.82		-1.33	0.184	0.72		1.07	0.287
	LSTM-Roll	-0.67		-1.06	0.288	0.87		1.32	0.187
Rolling β	Logistic	-1.14		-1.21	0.228	0.68		0.82	0.413
	RF	0.23		0.48	0.632	2.06	*	1.94	0.052
	XGB	-0.06		-0.10	0.922	1.77	*	1.76	0.078
	LSTM-Exp	-0.70		-0.90	0.370	1.13		1.12	0.264
	LSTM-Roll	-0.51		-0.64	0.520	1.31		1.34	0.182
EWMA β	Logistic	-0.78		-0.93	0.355	0.51		0.63	0.527
	RF	0.18		0.39	0.693	1.47		1.46	0.145
	XGB	-0.08		-0.14	0.892	1.21		1.28	0.200
	LSTM-Exp	-0.45		-0.71	0.479	0.84		0.81	0.418
	LSTM-Roll	-0.33		-0.46	0.643	0.96		0.99	0.324
Kalman β	Logistic	-0.96		-1.26	0.207	0.73		0.87	0.384
	RF	0.04		0.09	0.925	1.74	*	1.80	0.072
	XGB	-0.26		-0.44	0.659	1.44		1.62	0.106
	LSTM-Exp	-0.56		-0.86	0.392	1.14		1.19	0.233
	LSTM-Roll	-0.61		-0.88	0.380	1.09		1.19	0.233

Notes: This table reports incremental alphas ($\Delta\alpha$, annualized in % p.a.) for all 25 ML strategies (5 baskets $\times$ 5 models) against two alternative benchmarks: Buy&Hold Offensive (left panel) and Buy&Hold Defensive (right panel). Incremental alphas are computed by regressing the difference between ML and benchmark returns on the Fama–French 6-factor (FF6) model. t -statistics use Newey–West HAC standard errors (12 lags) on monthly data. p = raw p -value. All returns are net of transaction costs. This robustness check complements the main 50/50 benchmark analysis (Table 18). Sample: July 1980–June 2024. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

TABLE 20
Subperiod Incremental Alpha: Full Set of ML Strategies vs. 50/50 Benchmark (FF5/FF6/FF7, Q5, Net Returns)

D.6.3. Subperiod Incremental Alpha: Full Set of ML Strategies vs. 50/50 Benchmark.

Basket	Model	P1 (1980–1999)					P2 (2000–2009)					P3 (2010–2024)				
		FF5	FF6	FF7	Q5	FF5	FF6	FF7	Q5	FF5	FF6	FF7	FF5	FF6	FF7	Q5
Economic	Logistic	0.42	0.43	0.29	0.76	-0.26	-0.26	-0.31	-0.55	-0.40	-0.44	-0.50	-0.40	-0.44	-0.50	-0.47
	RF	-0.30	0.04	0.02	0.57	-1.43	**	-1.46	***	-0.02	-0.03	-0.04	-0.02	-0.03	-0.04	0.09
	XGB	0.36	0.63	0.51	0.77	-0.36	-0.35	-0.42	0.26	-0.30	-0.31	-0.26	-0.30	-0.31	-0.26	-0.26
	LSTM-Exp	-0.28	0.01	0.01	0.51	-0.15	-0.15	-0.25	0.04	-0.16	-0.19	-0.29	-0.16	-0.19	-0.29	-0.21
	LSTM-Roll	-0.21	-0.03	-0.00	0.40	-0.15	-0.15	-0.25	0.04	-0.58	-0.64	-0.75	-0.58	-0.64	-0.75	-0.61
Beta Exp.	Logistic	0.02	0.84	0.60	-0.19	-1.53	-1.53	-1.56	-1.47	0.52	0.59	0.49	0.52	0.59	0.49	0.51
	RF	0.89	1.33	***	0.23	2.24	**	2.27	**	0.31	0.40	0.29	0.31	0.40	0.29	0.15
	XGB	0.69	1.10	**	0.16	2.36	**	2.46	***	-0.62	-0.61	-0.78	-0.62	-0.61	-0.78	-0.86
	LSTM-Exp	0.66	1.23	***	0.51	-0.24	-0.27	-0.31	1.24	0.42	0.50	0.45	0.42	0.50	0.45	0.43
	LSTM-Roll	1.00	**	1.35	**	-0.24	-0.27	-0.31	-1.60	0.60	0.58	0.81	0.60	0.58	0.81	0.77
Rolling β	Logistic	-0.12	0.64	0.39	-0.57	-1.87	-1.86	-1.81	-1.40	0.78	0.83	0.58	0.78	0.83	0.58	0.64
	RF	0.90	1.18	*	-0.30	3.57	***	3.65	***	0.67	0.82	0.53	0.67	0.82	0.53	0.51
	XGB	1.12	*	1.44	**	3.04	**	3.25	***	-0.61	-0.53	-0.84	-0.61	-0.53	-0.84	-0.90
	LSTM-Exp	0.83	1.25	**	0.39	0.13	0.10	0.21	-1.53	0.67	0.73	0.46	0.67	0.73	0.46	0.54
	LSTM-Roll	1.50	**	2.05	***	0.13	0.10	0.21	-1.53	0.27	0.33	0.26	0.27	0.33	0.26	0.27
EWMA β	Logistic	0.04	0.59	0.35	-0.61	-1.86	-1.83	-1.76	-0.95	0.46	0.54	0.18	0.46	0.54	0.18	0.21
	RF	0.58	0.76	0.90	-0.71	3.66	***	3.77	***	0.12	0.29	-0.07	0.12	0.29	-0.07	-0.14
	XGB	0.92	1.07	1.05	-0.35	2.78	**	2.96	***	-1.13	-1.01	-1.39	-1.13	-1.01	-1.39	*
	LSTM-Exp	0.53	0.79	1.04	-0.09	0.24	0.23	0.36	-1.27	0.35	0.45	0.08	0.35	0.45	0.08	0.11
	LSTM-Roll	1.16	*	1.59	**	0.24	0.23	0.36	-1.27	-0.30	-0.26	-0.27	-0.30	-0.26	-0.27	-0.33
Kalman β	Logistic	-0.05	0.58	0.39	-0.53	-1.39	-1.37	-1.25	-0.93	0.78	0.80	0.50	0.78	0.80	0.50	0.52
	RF	0.70	0.98	1.06	-0.65	3.58	***	3.60	***	0.55	0.68	0.34	0.55	0.68	0.34	0.27
	XGB	1.11	*	1.37	**	2.82	**	2.95	***	-0.86	-0.80	-1.17	-0.86	-0.80	-1.17	-1.26
	LSTM-Exp	0.83	1.25	**	0.24	0.28	0.26	0.39	-1.26	0.71	0.74	0.43	0.71	0.74	0.43	0.48
	LSTM-Roll	1.28	**	1.61	**	0.28	0.26	0.39	-1.26	-0.08	-0.07	-0.06	-0.08	-0.07	-0.06	-0.12

Notes: This table reports subperiod incremental alphas ($\Delta\alpha$, annualized in % p.a.) for the full set of 25 ML regime-switching strategies (5 baskets $\times$ 5 models) relative to a fixed 50/50 benchmark (50% Buy&Hold Offensive + 50% Buy&Hold Defensive), using net-of-cost returns. For each subperiod $P1$ (1980–1999), $P2$ (2000–2009), and $P3$ (2010–2024), $\Delta\alpha$ is estimated from time-series regressions of (ML minus benchmark) returns on the Fama-French FF5/FF6/FF7 factor models and on the Hou-Xue-Zhang q -factor model (Q5). FF7 extends FF6 by adding the quality-minus-junk (QMJ) factor. Stars indicate statistical significance based on Newey–West HAC standard errors (12 lags) computed on monthly data. The set of strategies is fixed ex-ante (no re-optimization within subperiods). Sample: July 1980–June 2024. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

TABLE 21
Subperiod Robustness: All ML Strategies vs 50/50 Benchmark (FF6)

Basket	Model	P1: 1980–1999		P2: 2000–2009		P3: 2010–2024	
		Sharpe	$\Delta\alpha$	Sharpe	$\Delta\alpha$	Sharpe	$\Delta\alpha$
Economic	Logistic	0.90	0.43	0.54	-0.26	0.75	-0.44
	RF	0.87	0.04	0.57	-1.42 **	0.74	-0.03
	XGB	0.90	0.63	0.61	-0.35	0.74	-0.31
	LSTM-Exp	0.87	0.01	0.54	-0.15	0.75	-0.19
	LSTM-Roll	0.88	-0.03	0.54	-0.15	0.74	-0.64
Beta Exp.	Logistic	0.93	0.84	0.55	-1.53	0.80	0.59
	RF	0.92	1.33 ***	0.61	2.23 **	0.80	0.40
	XGB	0.92	1.10 **	0.52	2.36 **	0.77	-0.61
	LSTM-Exp	0.91	1.23 ***	0.67	-0.27	0.80	0.50
	LSTM-Roll	0.94	1.12 *	0.67	-0.27	0.80	0.58
Rolling β	Logistic	0.92	0.64	0.57	-1.86	0.78	0.83
	RF	0.89	1.18 *	0.65	3.56 ***	0.79	0.82
	XGB	0.90	1.44 **	0.53	3.06 ***	0.74	-0.53
	LSTM-Exp	0.88	1.25 **	0.70	0.10	0.78	0.73
	LSTM-Roll	0.94	1.73 **	0.70	0.10	0.77	0.33
EWMA β	Logistic	0.94	0.59	0.57	-1.83	0.77	0.54
	RF	0.88	0.76	0.63	3.65 ***	0.77	0.29
	XGB	0.91	1.07	0.49	2.79 **	0.73	-1.01
	LSTM-Exp	0.88	0.79	0.71	0.23	0.77	0.45
	LSTM-Roll	0.93	1.31 *	0.71	0.23	0.75	-0.26
Kalman β	Logistic	0.92	0.58	0.58	-1.37	0.78	0.80
	RF	0.88	0.98	0.65	3.56 ***	0.78	0.68
	XGB	0.91	1.37 **	0.52	2.84 ***	0.73	-0.80
	LSTM-Exp	0.89	1.25 **	0.70	0.26	0.78	0.74
	LSTM-Roll	0.93	1.36 **	0.70	0.26	0.76	-0.07

Notes: This table reports subperiod performance metrics for all 25 ML strategies (5 baskets $\times$ 5 models) relative to the 50/50 benchmark (50% Buy&Hold Offensive + 50% Buy&Hold Defensive). Sharpe denotes the net Sharpe ratio of the ML strategy within each subperiod. $\Delta\alpha$ denotes the incremental alpha (in % p.a.) from the FF6 regression of monthly return differentials (ML minus 50/50). Subperiods are: P1 (1980–1999), P2 (2000–2009), and P3 (2010–2024). Statistical significance stars are based on Newey–West HAC t -statistics with 12 lags (not reported). All returns are net of transaction costs. Sample: July 1980–June 2024. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

TABLE 22
Break-Even Transaction Cost Analysis: All ML Strategies

2*Basket	2*Model	2*NET Return (% p.a.)	Panel A: vs Market (14.71%)			Panel B: vs 50/50 Benchmark		
			Headroom (bps/yr)	Total TC @ BE (bps/yr)	TC Mult.	Headroom (bps/yr)	Total TC @ BE (bps/yr)	TC Mult.
Economic	LSTM-Exp	14.82	11	68	1.20	-77	0	0.00
	LSTM-Roll	14.82	11	67	1.19	-86	0	0.00
	Logistic	15.01	30	86	1.53	-65	0	0.00
	RF	15.05	34	90	1.60	-60	0	0.00
	XGBoost	15.40	69	126	2.23	-25	32	0.56
Beta Expansion	LSTM-Exp	16.25	154	219	3.34	-0	65	1.00
	LSTM-Roll	16.29	158	223	3.40	0	65	1.00
	Logistic	15.83	112	178	2.71	-62	3	0.05
	RF	16.14	143	209	3.18	6	71	1.09
	XGBoost	15.56	85	151	2.29	-10	55	0.85
Rolling Beta	LSTM-Exp	16.01	130	211	2.61	2	80	1.02
	LSTM-Roll	16.16	145	226	2.79	-50	28	0.36
	Logistic	15.61	90	171	2.11	-69	10	0.12
	RF	16.08	137	218	2.69	-13	65	0.83
	XGBoost	15.39	68	149	1.84	-50	28	0.36
EWMA Beta	LSTM-Exp	15.96	125	217	2.36	18	107	1.20
	LSTM-Roll	16.00	129	221	2.40	-29	60	0.68
	Logistic	15.72	101	193	2.10	-54	35	0.39
	RF	15.86	115	207	2.25	-12	77	0.86
	XGBoost	15.19	48	140	1.52	-33	56	0.63
Kalman Beta	LSTM-Exp	16.08	137	211	2.85	19	87	1.28
	LSTM-Roll	16.02	131	206	2.77	-27	41	0.61
	Logistic	15.68	97	171	2.31	-52	16	0.24
	RF	16.00	129	203	2.74	-1	67	0.98
	XGBoost	15.33	62	137	1.84	-20	48	0.71

Notes:

- The transaction-cost figures reported elsewhere reflect one-way stock trading costs only (June rebalance), whereas Real TC here includes both stock-level costs and inter-basket switching costs, hence higher annualized totals.
- **NET Return:** Annual return after real transaction costs (stock TC + inter-basket TC).
- **Real TC:** Actual transaction costs incurred (56-89 bps/year), including both stock-level rebalancing (June only) and inter-basket switching costs (monthly), computed using Frazzini et al. (2018) schedule.
- **Panel A (vs Market):** Break-even analysis vs NYSE Benchmark (14.71% Ann. Return, consistent with Table 1).
- **Panel B (vs 50/50):** Break-even analysis vs 50/50 Buy&Hold Offensive/Defensive benchmark, computed net using its own turnover-based TC model (annual June rebalance).
- **Headroom:** Additional TC (bps/year) that would eliminate excess return vs benchmark. Negative headroom indicates underperformance.
- **Total TC @ BE:** $\max(0, \text{Real TC} + \text{Headroom})$ — maximum sustainable TC level before strategy loses its edge.
- **TC Mult.:** Total TC @ BE / Real TC (how many times current TC can be sustained).
- When $\text{Real TC} + \text{Headroom} \leq 0$, Total TC @ BE is set to 0 and TC Mult. to 0.00 (strategy already underperforms the benchmark net of current costs).
- TC Multiplier > 1 indicates robustness; $= 0$ indicates the strategy does not beat the benchmark.

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